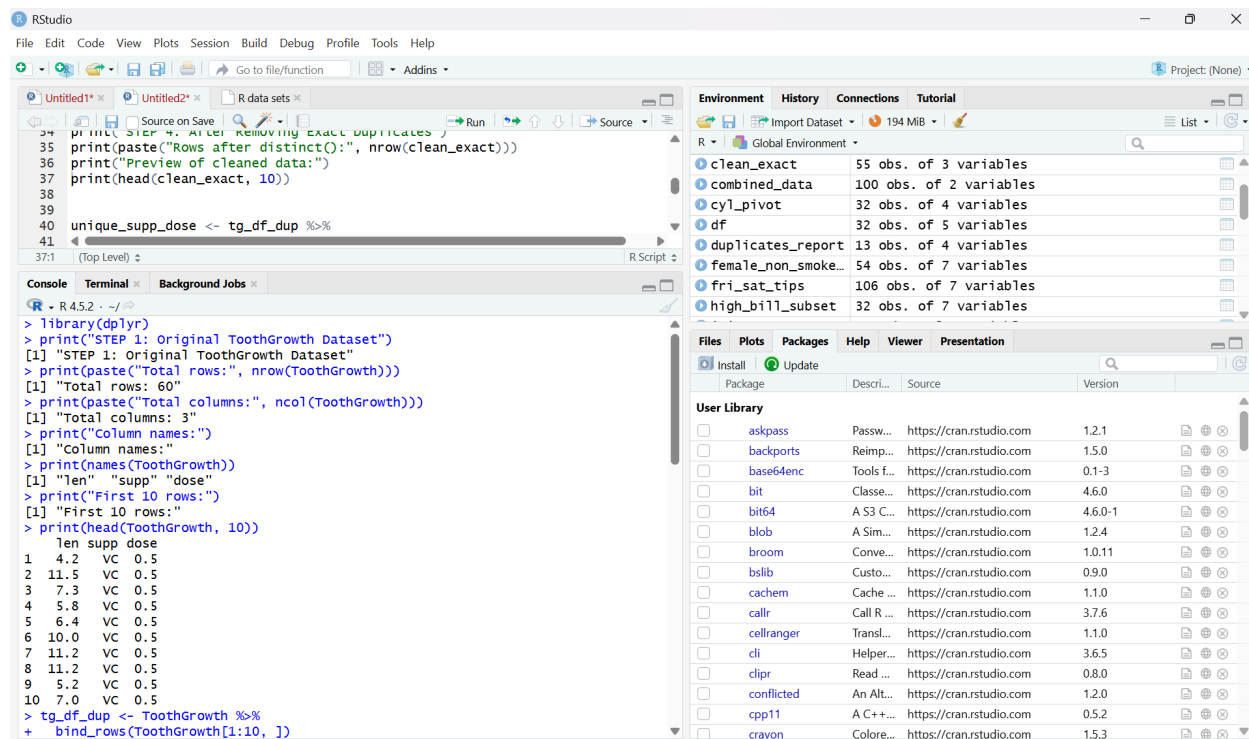


SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE

SUBJECT - Data Analysis with SAS / SPSS / R

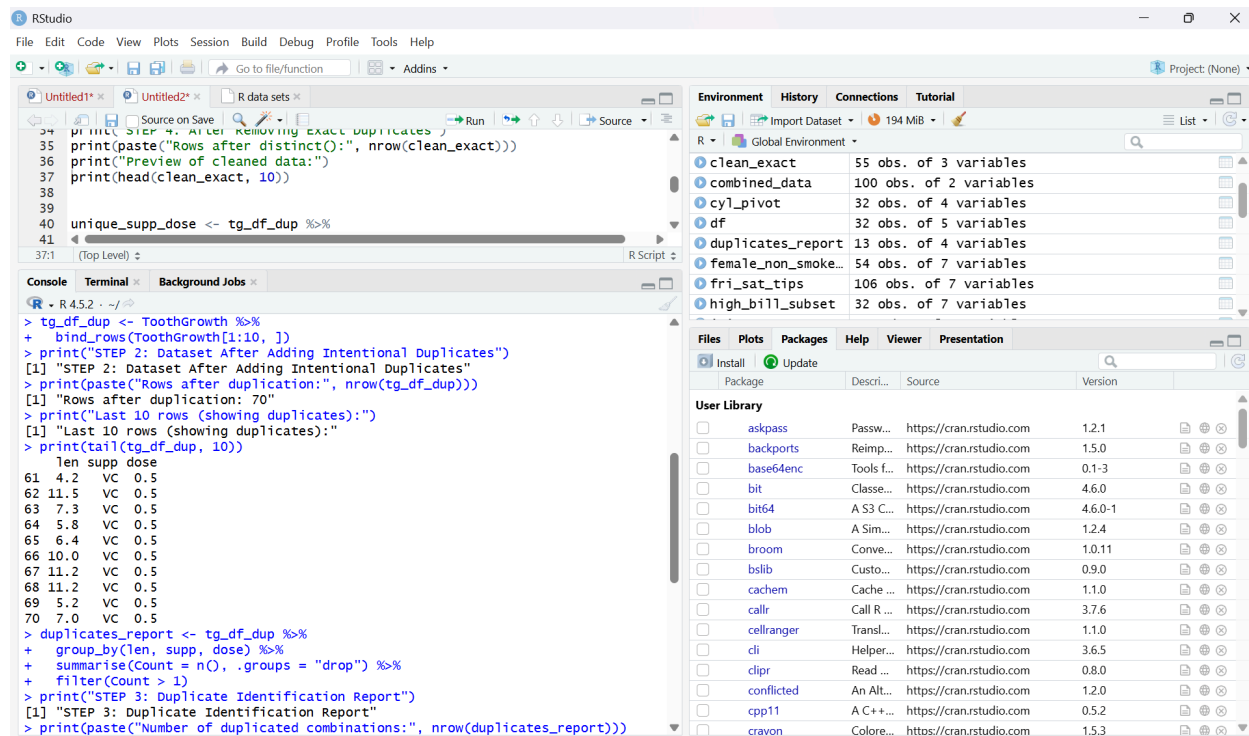
Aim :- Identifying and handling duplicates using distinct() (R studio).

Output :-



```
RStudio
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Go to file/function Addins
Project: (None)
Environment History Connections Tutorial
R Global Environment
clean_exact 55 obs. of 3 variables
combined_data 100 obs. of 2 variables
cyl_pivot 32 obs. of 4 variables
df 32 obs. of 5 variables
duplicates_report 13 obs. of 4 variables
female_non_smoke... 54 obs. of 7 variables
fri_sat_tips 106 obs. of 7 variables
high_bill_subset 32 obs. of 7 variables
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backports Reimp... https://cran.rstudio.com 1.5.0
base64enc Tools f... https://cran.rstudio.com 0.1-3
bit Classe... https://cran.rstudio.com 4.6.0
bit64 A S3 C... https://cran.rstudio.com 4.6.0-1
blob A Sim... https://cran.rstudio.com 1.2.4
broom Conve... https://cran.rstudio.com 1.0.11
bslib Cust... https://cran.rstudio.com 0.9.0
cachem Cache ... https://cran.rstudio.com 1.1.0
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clipr Read ... https://cran.rstudio.com 0.8.0
conflicted An Alt... https://cran.rstudio.com 1.2.0
cpp11 A C++... https://cran.rstudio.com 0.5.2
cravon Colore... https://cran.rstudio.com 1.5.3

> library(dplyr)
> print("STEP 1: Original ToothGrowth Dataset")
[1] "STEP 1: Original ToothGrowth Dataset"
> print(paste("Total rows:", nrow(ToothGrowth)))
[1] "Total rows: 60"
> print(paste("Total columns:", ncol(ToothGrowth)))
[1] "Total columns: 3"
> print("Column names:")
[1] "len" "supp" "dose"
> print(names(ToothGrowth))
[1] "len" "supp" "dose"
> print("First 10 rows:")
[1] "First 10 rows:"
> print(head(ToothGrowth, 10))
  len supp dose
1  4.2 VC 0.5
2 11.5 VC 0.5
3  7.3 VC 0.5
4  5.8 VC 0.5
5  6.4 VC 0.5
6 10.0 VC 0.5
7 11.2 VC 0.5
8 11.2 VC 0.5
9  5.2 VC 0.5
10 7.0 VC 0.5
> tg_df_dup <- ToothGrowth %>%
+   bind_rows(ToothGrowth[1:10, ])
> print("STEP 4: After Removing exact Duplicates")
[1] "STEP 4: After Removing exact Duplicates"
> print(paste("Rows after distinct():", nrow(clean_exact)))
[1] "Rows after distinct(): 55"
> print(head(clean_exact, 10))
  len supp dose
1  4.2 VC 0.5
2 11.5 VC 0.5
3  7.3 VC 0.5
4  5.8 VC 0.5
5  6.4 VC 0.5
6 10.0 VC 0.5
7 11.2 VC 0.5
8 11.2 VC 0.5
9  5.2 VC 0.5
10 7.0 VC 0.5
```



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cpp11 A C++... https://cran.rstudio.com 0.5.2
cravon Colore... https://cran.rstudio.com 1.5.3

> tg_df_dup <- ToothGrowth %>%
+   bind_rows(ToothGrowth[1:10, ])
> print("STEP 2: Dataset After Adding Intentional Duplicates")
[1] "STEP 2: Dataset After Adding Intentional Duplicates"
> print(paste("Rows after duplication:", nrow(tg_df_dup)))
[1] "Rows after duplication: 70"
> print("Last 10 rows (showing duplicates):")
[1] "Last 10 rows (showing duplicates):"
> print(tail(tg_df_dup, 10))
  len supp dose
61  4.2 VC 0.5
62 11.5 VC 0.5
63  7.3 VC 0.5
64  5.8 VC 0.5
65  6.4 VC 0.5
66 10.0 VC 0.5
67 11.2 VC 0.5
68 11.2 VC 0.5
69  5.2 VC 0.5
70  7.0 VC 0.5
> duplicates_report <- tg_df_dup %>%
+   group_by(len, supp, dose) %>%
+   summarise(count = n(), .groups = "drop") %>%
+   filter(count > 1)
> print("STEP 3: Duplicate Identification Report")
[1] "STEP 3: Duplicate Identification Report"
> print(paste("Number of duplicated combinations:", nrow(duplicates_report)))
[1] "Number of duplicated combinations: 13"
```

Name - Siddhant Chalke
Roll No - S073

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```
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cravon Colore... https://cran.rstudio.com 1.5.3

Untitled1* R data sets
34 print("STEP 4: After Removing Exact Duplicates")
35 print(paste("Rows after distinct():", nrow(clean_exact)))
36 print("Preview of cleaned data:")
37 print(head(clean_exact, 10))
38
39
40 unique_supp_dose <- tg_df_dup %>%
41
37:1 (Top Level) R Script

Console Terminal Background Jobs
R - R 4.5.2 - /
70 7.0 VC 0.5
> duplicates_report <- tg_df_dup %>%
+ group_by(len, supp, dose) %>%
+ summarise(count = n(), .groups = "drop") %>%
+ filter(count > 1)
> print("STEP 3: Duplicate Identification Report")
[1] "STEP 3: Duplicate Identification Report"
> print(paste("Number of duplicated combinations:", nrow(duplicates_report)))
[1] "Number of duplicated combinations: 13"
> print(duplicates_report)
# A tibble: 13 x 4
  len supp dose count
  <dbl> <fct> <dbl> <int>
1 4.2 VC 0.5 2
2 5.2 VC 0.5 2
3 5.8 VC 0.5 2
4 6.4 VC 0.5 2
5 7 VC 0.5 2
6 7.3 VC 0.5 2
7 9.7 OJ 0.5 2
8 10 VC 0.5 2
9 11.2 VC 0.5 4
10 11.5 VC 0.5 2
11 16.5 VC 1 2
12 17.3 VC 1 2
13 26.4 OJ 2 2
> clean_exact <- tg_df_dup %>%
```

```
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Untitled1* R data sets
34 print("STEP 4: After Removing Exact Duplicates")
35 print(paste("Rows after distinct():", nrow(clean_exact)))
36 print("Preview of cleaned data:")
37 print(head(clean_exact, 10))
38
39
40 unique_supp_dose <- tg_df_dup %>%
41
37:1 (Top Level) R Script

Console Terminal Background Jobs
R - R 4.5.2 - /
> print(duplicates_report)
# A tibble: 13 x 4
  len supp dose count
  <dbl> <fct> <dbl> <int>
1 4.2 VC 0.5 2
2 5.2 VC 0.5 2
3 5.8 VC 0.5 2
4 6.4 VC 0.5 2
5 7 VC 0.5 2
6 7.3 VC 0.5 2
7 9.7 OJ 0.5 2
8 10 VC 0.5 2
9 11.2 VC 0.5 4
10 11.5 VC 0.5 2
11 16.5 VC 1 2
12 17.3 VC 1 2
13 26.4 OJ 2 2
> clean_exact <- tg_df_dup %>%
+ distinct()
> print("STEP 4: After Removing Exact Duplicates")
[1] "STEP 4: After Removing Exact Duplicates"
> print(paste("Rows after distinct():", nrow(clean_exact)))
[1] "Rows after distinct(): 55"
> print("Preview of cleaned data:")
[1] "Preview of cleaned data:"
>
```

Name - Siddhant Chalke
Roll No - S073